Software Requirement Specification (SRS)

Project: Cardiac Diagnosis System

Problem Statement

The Cardiac Diagnosis System is designed to provide a scalable and efficient platform for managing and analyzing cardiac diagnosis data. In many healthcare environments, patient diagnosis data is scattered across systems and lacks centralized access for analysis and decision-making. Healthcare professionals require a system that can retrieve diagnosis information, analyze treatment recommendations, and allow secure access to personalized data.

To address this need, the Cardiac Diagnosis System will be developed using a microservices architecture. The system will integrate with an external Diagnosis API, allow users to search and analyze cardiac-related information, and provide secure authentication and personalized bookmarking features.The Cardiac Diagnosis System offers a scalable and efficient platform for managing and analyzing cardiac diagnosis data, addressing the current challenge of scattered patient information. Healthcare professionals need centralized access to retrieve diagnosis data, analyze treatment recommendations, and securely access personalized information.

To achieve this, the system will use a microservices architecture, integrating with an external Diagnosis API. Key features will include searching and analyzing cardiac information, secure authentication, and personalized bookmarking.

Features

- ➢ User authentication will use JWT token
- ➢ Accessible to registered users and guest users

- ➢ Registered users will be able to perform the following activities:
    - Register with the application
    - Login and logout securely
    - Search diagnosis data by pain type, age, blood pressure, and gender
    - Bookmark diagnosis records
    - View bookmarked records

- Delete bookmarked records
- Update personal information
- Reset password

➢ Guest users will be able to perform the following activities:
➢ View limited diagnosis data
➢ Perform basic search

Components

General Components

• Home
Display system overview and cardiac diagnosis information

• Header
Allows users to navigate through pages, search, and logout

• Footer
Provides contact and support information

• Signup
Register to the application by providing personal details

• Login
Login to the application using username and password

User Components

• **View Diagnosis**
Display list of diagnosis records retrieved from external API
When clicked, display detailed information
Provide filter/search options

• **Bookmark Section**
Display list of bookmarked diagnosis records
Provide option to delete bookmark

- **Profile**
  Display personal user information
  Allow updating of details
  Allow password reset

Microservices

Authentication Service

Registers and authenticates users and provides JWT token

- /register
- /login
- /validateToken
- /logout

UserProfile Service

Provides services for managing user personal details

- /createProfile
- /viewProfile/{userId}
- /updateProfile/{userFields**}
- /deleteProfile/{userId}
- /resetPassword/{userId}

Diagnosis Service

Provides services for managing diagnosis data

- /getAllDiagnosis
- /getDiagnosisById/{diagnosisId}
- /searchDiagnosis
- /filterDiagnosis

Bookmark Service

Provides services for bookmarked records

- /addBookmark
- /viewBookmarks/{userId}
- /deleteBookmark/{bookmarkId}

Models

User

- username : String (@Id)
- password : String
- role : String

UserProfile

- userId : String (@Id)
- firstName : String
- lastName : String
- email : String
- phoneNumber : Number
- gender : String
- age : Number

Bookmark

- bookmarkId : String (@Id)
- userId : String
- diagnosisId : String
- bookmarkedDate : LocalDateTime

External Diagnosis API

Docker Command
docker run -p3232:3232 --name diagnocontainer stackroutenew/diagnosisapi

Access URL
http://localhost:3232/diagnosis

Available Endpoints

- GET /diagnosis
- GET /diagnosis?property=value
- POST /diagnosis